

YouTube API Services - Recommendation & Notification Architecture

1.9.0 OAS 3.0

Base URL

Type	URL
All Traffic	https://api.youtube.com/api
Internal endpoints are prefixed with /internal/ and are blocked from external access at the API Gateway level.	

Module Overview

#	Module	Submodules
1	Unified Feed Generation & Content Discovery	Auth & Sessions, Videos & Feeds
2	Watch History Analysis & Behavior Tracking	Video Playback & Streaming, Settings & Users
3	User Preference & Personalization Control	Engagement & Interactions, Playlists, Subscriptions, Inter-Service Data Services
4	Notification System	Notifications, Inter-Service Event Triggers

Authentication

All protected endpoints require a Bearer JWT token obtained from POST /auth/google or POST /auth/refresh .

Servers

https://api.youtube.com/api - API Gateway (routes /internal/* to microservices internally)

Authorize

Module: Unified Feed Generation & Content Discovery

Module 1 ▸ Auth & Sessions

User authentication via Google OAuth 2.0, JWT token lifecycle (issue, refresh, revoke), and session initialization — capturing geo, device, and browser context used downstream by the Recommendation Engine to personalize the feed.

POST /auth/google Authenticate via Google Account

Exchanges a Google OAuth 2.0 ID token or authorization code for a YouTube API session. Acts as both login and registration (if the YouTube profile doesn't exist yet).

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{  "idToken": "eyJhbGciOiJSUzI1NiIsImtpZCI6..."}
```

Responses

Code	Description	Links
200	Successfully authenticated; YouTube profile provisioned. Media type application/json Controls Accept header. Example Value Schema <pre>{ "userId": "user_abc123", "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "refreshToken": "ref_token_xyz789_abcdef", "expiresIn": 3600}</pre>	No links
401	Unauthorized — missing or invalid Bearer token. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links

POST

/auth/refresh

Refresh JWT Token

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Exchanges a valid refresh token for a new short-lived JWT access token.

Parameters

Try it out

No parameters

Request body

required

application/json

⌵

Example Value

Schema

```
{  "refreshToken": "ref_token_xyz789..."}
```

Responses

Code	Description	Links
200	Tokens successfully refreshed. Media type application/json Controls Accept header. Example Value Schema <pre>{ "userId": "user_abc123", "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "refreshToken": "ref_token_xyz789_abcdef", "expiresIn": 3600}</pre>	

 No links || 401 | Unauthorized — missing or invalid Bearer token. Media type | No links |

Code	Description	Links
application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>		

POST

/auth/logout Logout User



Invalidates the current session and revokes the active refresh token.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
204	Successfully logged out.	No links
401	Unauthorized — missing or invalid Bearer token.	No links

Media type

application/json

Example Value | Schema

```
{  "error": {    "code": 400,    "message": "Invalid Request"  } }
```

GET

/auth/profile

Get Current User Profile



Retrieves the core identity profile of the currently authenticated user.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Successfully retrieved user profile. Media type application/json Controls Accept header. Example Value Schema	No links

Code	Description	Links
	<pre>{ "userId": "user_abc123", "email": "user@gmail.com", "displayName": "Tech Enthusiast", "avatarUrl": "https://youtube.com/img/avatar.jpg", "createdAt": "2020-05-15T14:30:00Z" }</pre>	
401	Unauthorized — missing or invalid Bearer token. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	No links

POST	/sessions Initialize Tracking Session	 
Called when the app opens. Generates an interaction session ID and extracts contextual data (geo, device, browser) used later by the Recommendation Engine.		
Parameters		Try it out
No parameters		
Request body required		application/json
Example Value Schema <pre>{ "deviceType": "iOS", "appVersion": "16.42.3" }</pre>		
Responses		
Code	Description	Links
201	Session created with context tracking active. Media type application/json Controls Accept header. Example Value Schema <pre>{ "sessionId": "sess_987654321", "geoContext": "US-NY" }</pre>	No links

Module: Unified Feed Generation & Content Discovery

Module 1 ▶ Videos & Feeds

Paginated endless-scroll home feed (delegates ranking to the AI Engine), individual video metadata loading, related/up-next recommendations, recommendation strategy control, and the internal raw video candidate endpoint consumed by the AI ranking models.

GET	/videos List Videos (Home Feed)	 

Generates the dynamic Home Feed catalog, optimized for endless scrolling. Delegates ranking to the Recommendation Engine based on the user's active session. Branches internally into a **Cold Start** strategy (new users) or **Personalization** strategy (returning users).

Parameters

Try it out

Name	Description
sessionId string (query)	sessionId
quantity integer (query)	Default value : 15 15
pageToken string (query)	pageToken

Responses

Code	Description	Links
200	Paginated list of ranked video candidates. Media type application/json Controls Accept header. Example Value Schema <pre>{ "nextPageToken": "CAUQAA", "pageInfo": { "totalResults": 1000, "resultsPerPage": 10 }, "items": [{ "id": "dQw4w9WgXcQ", "snippet": { "channelId": "UC38IQsAvIsxxjztdMZQtwHA", "title": "Never Gonna Give You Up", "durationSeconds": 212 }, "statistics": { "viewCount": 1483929411, "trendingScore": 98.4 } }] }</pre>	No links
429	Too Many Requests — rate limit exceeded. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	No links

GET /videos/{videoId} Load Video Metadata

Fetches specific video metadata, description, and statistics necessary prior to starting playback.

Parameters

Try it out

Name	Description
videoid * required string (path)	videoid

Responses

Code	Description	Links
200	Video details loaded successfully. Media type application/json Controls Accept header. Example Value Schema <pre>{ "id": "dQw4w9WgXcQ", "snippet": { "channelId": "UC38IQsAvIsxxjztdMZQtwHA", "title": "Never Gonna Give You Up", "durationSeconds": 212 }, "statistics": { "viewCount": 1483929411, "trendingScore": 98.4 } }</pre>	No links
404	Not Found — resource does not exist. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	No links

GET

/videos/{videoId}/related

Get Related Videos (Up Next)

🔒 ⤴

Generates the sidebar/up-next recommendation feed based on the currently watching video and user context.

Parameters

Try it out

Name	Description
videoid * required string (path)	videoid

Responses

Code	Description	Links
200	Ranked list of related candidates.	No links

Code	Description	Links
	Media type application/json Controls Accept header. Example Value Schema <pre>{ "nextPageToken": "CAUQAA", "pageInfo": { "totalResults": 1000, "resultsPerPage": 10 }, "items": [{ "id": "dQw4w9WgXcQ", "snippet": { "channelId": "UC38IQsAvIsxxjztdMZQtwHA", "title": "Never Gonna Give You Up", "durationSeconds": 212 }, "statistics": { "viewCount": 1483929411, "trendingScore": 98.4 } }]}</pre>	

PUT	/internal/recommendations-strategy [INTERNAL] Set Recommendation Strategy	 
Configures the active recommendation strategy for the feed engine. COLD_START — new/anonymous users; falls back to trending/regional content. PERSONALIZED — returning users; uses watch history, preference vector, and session context.		
Parameters No parameters		Try it out
Request body required		application/json
Example Value Schema <pre>{ "strategy": "PERSONALIZED", "userId": "user_abc123", "sessionId": "sess_987654321"}</pre>		
Responses		
Code	Description	Links
200	Strategy successfully applied. Media type application/json Controls Accept header. Example Value Schema <pre>{ "strategy": "PERSONALIZED", "appliedAt": "2025-03-01T10:00:00Z"}</pre>	No links
400	Bad Request — invalid parameters or payload. Media type application/json Example Value Schema	No links

Code	Description	Links
	<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	
401	Unauthorized — missing or invalid Bearer token.	No links
Media type		
application/json		
Example Value Schema		
	<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	

GET

/internal/videos/candidates [INTERNAL] Fetch Raw Video Catalog (AI Engine → Data Services)

Fetches raw video catalogs and trending scores so the AI Engine can run its ranking models.

Parameters

Try it out

Name	Description
region string (query)	region
limit integer (query)	Default value : 500 500

Responses

Code	Description	Links
200	Raw candidate list.	No links
Media type		
application/json		
Controls Accept header.		
Example Value Schema		
	<pre>{ "nextPageToken": "CAUQAA", "pageInfo": { "totalResults": 1000, "resultsPerPage": 10 }, "items": [{ "id": "dQw4w9WgXcQ", "snippet": { "channelId": "UC38IQsAvIsxxjztdMZQtwHA", "title": "Never Gonna Give You Up", "durationSeconds": 212 }, "statistics": { "viewCount": 1483929411, "trendingScore": 98.4 } }] }</pre>	

Module 2 ▶ Video Playback & Streaming

Stream manifest retrieval, video chapter markers, watch-start event signaling, real-time playback telemetry (PLAY / PAUSE / SEEK / BUFFER), cross-device position sync, and watch summary submission to the Analytics Service.



GET

/videos/{videoId}/stream

Get Stream Manifest

Retrieves the HLS or DASH manifest URLs, available resolutions, and DRM licenses required for the video player to fetch media chunks.

Parameters

Try it out

Name	Description
videoId ★ required	
string (path)	videoId

Responses

Code	Description	Links
200	Stream manifest URLs successfully retrieved. Media type application/json Controls Accept header. Example Value Schema <pre>{ "hlsManifestUrl": "https://manifest.youtube.com/api/hls/playlist.m3u8", "resolutions": ["1080p", "720p", "480p", "360p"]}</pre>	No links
404	Not Found — resource does not exist. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links

GET

/videos/{videoId}/chapters

Get Video Chapters

Retrieves timeline markers and chapters for the frontend player UI.

Parameters

Try it out

Name	Description
videoid * required string (path)	videoid

Responses

Code	Description	Links
200	Chapters retrieved successfully. Media type application/json Controls Accept header. Example Value Schema <pre>{ "chapters": [{ "title": "Introduction", "startTimestamp": 0 }]}</pre>	No links

POST	/videos/{videoId}/playback/start	Send Watch Start	
Triggered immediately when the player successfully loads the first frame. Opens the session record in Data Services.			
Parameters			Try it out
Name	Description		
videoid * required string (path)	videoid		
Request body required			application/json
Example Value Schema <pre>{ "sessionId": "sess_987654321"}</pre>			
Responses			
Code	Description	Links	
202	Watch start registered.	No links	

POST	/videos/playback/telemetry	Record Playback Telemetry	

Aggregates real-time events (**PLAY** , **PAUSE** , **SEEK** , **BUFFER**) while the user is watching.

Parameters

[Try it out](#)

No parameters

Request body **required**

application/json ▾

Example Value | Schema

```
{
  "videoId": "dQw4w9WgXcQ",
  "sessionId": "sess_987654321",
  "events": [
    {
      "eventType": "PLAY",
      "timestamp": 1699564800000
    }
  ]
}
```

Responses

Code	Description	Links
202	Telemetry events accepted into the aggregation queue.	No links
400	Bad Request — invalid parameters or payload.	No links

Media type

application/json ▾

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

POST /videos/{videoId}/watch-summary Submit Watch Summary



Called at end of session. Sends aggregated watch data (total minutes, skip ratio) to Analytics, which then updates Data Services.

Parameters

[Try it out](#)

Name	Description
------	-------------

videoid * **required**

string
(path)

videoid

Request body **required**

application/json ▾

Example Value | Schema

```
{
  "sessionId": "sess_987654321",
  "watchMinutes": 3.5,
  "skipRatio": 0.15
}
```

Responses		
Code	Description	Links
202	Watch summary accepted and queued for processing.	No links

PUT

/videos/{videoId}/playback/position Sync Playback Position

🔒 ⬆

Saves the user's current playback timestamp to enable the 'Resume Watching' feature across devices.

Parameters

Try it out

Name	Description
videoId * required string (path)	videoId

Request body required

application/json ▼

Example Value | Schema

```
{  "positionSeconds": 134}
```

Responses

Code	Description	Links
204	Playback position successfully synced.	No links

Module 2 ▶ Settings & Users

Module: Watch History Analysis & Behavior Tracking

⬆

Watch history retrieval and deletion, user settings (privacy toggles, preference vector), and the internal endpoint for the Analytics Service to append watch summaries to the user's permanent database record.

GET

/history Retrieve Watch History

🔒 ⬆

Fetches the user's previously watched videos, ordered chronologically.

Parameters

Try it out

Name	Description
pageToken string (query)	pageToken

Responses

Code	Description	Links
200	Successfully retrieved watch history.	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "nextPageToken": "CAUQAA",
  "pageInfo": {
    "totalResults": 1000,
    "resultsPerPage": 10
  },
  "items": [
    {
      "id": "dQw4w9WgXcQ",
      "snippet": {
        "channelId": "UC38IQsAvIsxxjztdMZQtwHA",
        "title": "Never Gonna Give You Up",
        "durationSeconds": 212
      },
      "statistics": {
        "viewCount": 1483929411,
        "trendingScore": 98.4
      }
    }
  ]
}
```

DELETE /history Clear Watch History



Triggers the deleteHistory() cascade, permanently removing all watch history entries.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
204	History successfully cleared.	No links

GET /settings Retrieve User Controls



Fetches all user settings including privacy , prefVector , and restrictions .

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	User settings payload.	No links

Media type

Code	Description	Links
	application/json Controls Accept header. Example Value Schema <pre>{ "privacy": { "dataTrackingEnabled": true }, "prefVector": { "interests": ["music", "gaming"] }}</pre>	

PUT	/settings/privacy	Update Privacy Settings	 
Parameters			
No parameters			
Request body		required	application/json
Example Value Schema			
<pre>{ "dataTrackingEnabled": false}</pre>			
Responses			
Code	Description	Links	
200	Privacy settings successfully updated.	No links	

PUT	/settings/preferences	Modify Preference Vector	 
Manually tunes the user's algorithmic topical interest tags.			
Parameters			
No parameters			
Request body		required	application/json
Example Value Schema			
<pre>{ "interests": ["technology", "music", "gaming"]}</pre>			
Responses			

Code	Description	Links
200	Preference vector successfully updated.	No links

POST

/internal/users/{userId}/history [INTERNAL] Append Watch History (Analytics → Data Services)

⌵

The Analytics Service appends the computed WatchSummary to the user's permanent database record.

Parameters

Try it out

Name	Description
userId * required string (path)	userId

Request body required

application/json ▾

Example Value | Schema

```
{
  "sessionId": "sess_987654321",
  "watchMinutes": 3.5,
  "skipRatio": 0.15
}
```

Responses

Code	Description	Links
201	History record appended.	No links

Module: User Preference & Personalization Control

Module 3 ▸ Engagement & Interactions

Explicit video interactions (LIKE, DISLIKE, CLIP, SAVE), video statistics, comments (read/write), share event logging, and algorithmic feed-correction feedback signals (NOT_INTERESTED, DONT_RECOMMEND_CHANNEL).

GET

/videos/{videoId}/stats Get Video Statistics

⌵

Retrieves detailed interaction statistics — views, likes, dislikes, comments, shares.

Parameters

Try it out

Name	Description
videoId * required string (path)	videoId

Responses

Code	Description	Links
200	Video statistics retrieved successfully. Media type application/json Controls Accept header. Example Value Schema <pre>{ "videoId": "dQw4w9WgXcQ", "likeCount": 16000000, "dislikeCount": 12000, "commentCount": 54302, "shareCount": 8500}</pre>	No links
404	Not Found — resource does not exist. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links

POST	/videos/{videoId}/interact	Interact with Video (Like / Dislike / Clip / Save)	 
Performs an explicit interaction on a video — LIKE, DISLIKE, CLIP, or SAVE.			
Parameters			Try it out
Name	Description		
videoid ★ required	string (path) videoid		
Request body required			application/json
Example Value Schema			
<pre>{ "action": "SAVE", "playlistId": "PL1234567890", "clipSettings": { "startTimes": 15000, "endTimeMs": 45000 }}</pre>			
Responses			
Code	Description	Links	
200	Interaction recorded successfully.	No links	
400	Bad Request — invalid parameters or payload. Media type application/json Example Value Schema	No links	

Code	Description	Links
	<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	
401	Unauthorized — missing or invalid Bearer token.	No links
	Media type application/json	
	Example Value Schema	
	<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	

POST /videos/{videoId}/feedback Submit Algorithmic Feedback

Handles feed-correction signals. Logged via `FeedbackManager` and triggers internal preference updates.

Parameters Try it out

Name	Description
videoid * required string (path)	videoid

Request body required application/json

Example Value | Schema

```
{
  "action": "NOT_INTERESTED"
}
```

Responses

Code	Description	Links
204	Algorithmic feedback successfully applied.	No links
400	Bad Request — invalid parameters or payload.	No links
	Media type application/json	
	Example Value Schema	
	<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	

GET /videos/{videoId}/comments List Video Comments

Retrieves a paginated list of top-level comments for the specified video.

Parameters

Try it out

Name	Description
------	-------------

pageToken string (query)	
--------------------------------	--

maxResults integer (query)	
----------------------------------	--

Default value : 20

videoid * required string (path)	
--	--

Responses

Code	Description	Links
------	-------------	-------

200	Paginated list of comments.	
-----	-----------------------------	--

No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "nextPageToken": "CAIQAA",
  "items": [
    {
      "commentId": "cmnt_123abc",
      "authorDisplayName": "Tech Enthusiast",
      "text": "This video was incredibly helpful, thanks for sharing!",
      "publishedAt": "2023-10-24T23:22:28Z",
      "likeCount": 42
    }
  ]
}
```

POST

/videos/{videoId}/comments Add a Comment



Posts a new top-level comment to the video.

Parameters

Try it out

Name	Description
------	-------------

videoid * required string (path)	
--	--

Request body required

application/json

Example Value | Schema

```
{
  "text": "This video was incredibly helpful, thanks for sharing!"
}
```

Responses

Code	Description	Links
201	Comment successfully posted.	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "commentId": "cmnt_123abc",
  "authorDisplayName": "Tech Enthusiast",
  "text": "This video was incredibly helpful, thanks for sharing!",
  "publishedAt": "2023-10-24T23:22:28Z",
  "likeCount": 42
}
```

401	Unauthorized — missing or invalid Bearer token.	No links
-----	---	----------

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

POST

/videos/{videoId}/share Record Share Interaction



Logs that the user shared this video. Provides a significant boost to the internal trending score.

Parameters

Try it out

NameDescription

videoid * required
string
(path)

videoid

Request body required

application/json

Example Value | Schema

```
{
  "platform": "twitter"
}
```

Responses

Code	Description	Links
202	Share interaction recorded for analytics.	No links

GET

/playlists

List Playlists



Returns all playlists owned by the authenticated user, or specific playlists by ID.

Parameters

Try it out

Name	Description
id string (query)	Comma-separated list of playlist IDs to retrieve. id
mine boolean (query)	Set to true to retrieve the authenticated user's playlists. --
pageToken string (query)	pageToken

Responses

Code	Description	Links
200	Successfully retrieved playlists. Media type application/json Controls Accept header. Example Value Schema <pre>{ "nextPageToken": "CAIQAA", "pageInfo": { "totalResults": 1000, "resultsPerPage": 10 }, "items": [{ "id": "PL1234567890", "snippet": { "title": "My Favorite Music", "description": "A collection of the best tracks." }, "status": { "privacyStatus": "public" } }]}</pre>	No links
401	Unauthorized — missing or invalid Bearer token. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links

POST

/playlists

Create Playlist



Creates a new playlist for the authenticated user.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "id": "PL1234567890",
  "snippet": {
    "title": "My Favorite Music",
    "description": "A collection of the best tracks."
  },
  "status": {
    "privacyStatus": "public"
  }
}
```

Responses

Code	Description	Links
201	Playlist successfully created.	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "id": "PL1234567890",
  "snippet": {
    "title": "My Favorite Music",
    "description": "A collection of the best tracks."
  },
  "status": {
    "privacyStatus": "public"
  }
}
```

400	Bad Request — invalid parameters or payload.	No links
-----	--	----------

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

401	Unauthorized — missing or invalid Bearer token.	No links
-----	---	----------

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

GET

/playlists/{playlistId} Get Playlist



Retrieves a single playlist by ID including its metadata.

Parameters

Try it out

Name

Description

playlistId * required

The unique ID of the playlist.

string
(path)

playlistId

Responses

Code

Description

Links

200

Playlist retrieved successfully.

No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "id": "PL1234567890",
  "snippet": {
    "title": "My Favorite Music",
    "description": "A collection of the best tracks."
  },
  "status": {
    "privacyStatus": "public"
  }
}
```

401

Unauthorized — missing or invalid Bearer token.

No links

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

404

Not Found — resource does not exist.

No links

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

PUT

/playlists/{playlistId} Update Playlist



Modifies a playlist's title, description, or privacy status.

Parameters

Try it out

Name

Description

playlistId * required

The unique ID of the playlist.

string

playlistId

(path)

Request body required

application/json

Example Value | Schema

```
{
  "id": "PL1234567890",
  "snippet": {
    "title": "My Favorite Music",
    "description": "A collection of the best tracks."
  },
  "status": {
    "privacyStatus": "public"
  }
}
```

Responses

Code	Description	Links
200	Playlist successfully updated. Media type application/json Controls Accept header. Example Value Schema <pre>{ "id": "PL1234567890", "snippet": { "title": "My Favorite Music", "description": "A collection of the best tracks." }, "status": { "privacyStatus": "public" } }</pre>	

 No links || 401 | Unauthorized — missing or invalid Bearer token. Media type application/json Example Value | Schema ``` { "error": { "code": 400, "message": "Invalid Request" } } ``` |

 No links || 404 | Not Found — resource does not exist. Media type application/json Example Value | Schema ``` { "error": { "code": 400, "message": "Invalid Request" } } ``` |

DELETE

/playlists/{playlistId}

Delete Playlist

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Permanently deletes the specified playlist.

Parameters

Try it out

Name

Description

playlistId * required
string
(path)

The unique ID of the playlist.

playlistId

Responses

Code

Description

Links

204

Playlist successfully deleted.

No links

401

Unauthorized — missing or invalid Bearer token.

No links

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

404

Not Found — resource does not exist.

No links

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

GET

/playlists/{playlistId}/items List Playlist Items



Returns all videos inside the specified playlist.

Parameters

Try it out

Name

Description

pageToken
string
(query)

pageToken

maxResults
integer
(query)

Default value : 20

20

playlistId * required
string
(path)

The unique ID of the playlist.

playlistId

Responses

Code	Description	Links
200	Successfully retrieved playlist items.	No links

Media type

application/json ▾

Controls Accept header.

Example Value | Schema

```
{
  "nextPageToken": "CAIQAA",
  "pageInfo": {
    "totalResults": 1000,
    "resultsPerPage": 10
  },
  "items": [
    {
      "id": "pli_abc123",
      "snippet": {
        "playlistId": "PL1234567890",
        "position": 0,
        "resourceId": {
          "videoId": "dQw4w9WgXcQ"
        }
      }
    }
  ]
}
```

404	Not Found — resource does not exist.	No links
-----	--------------------------------------	----------

Media type

application/json ▾

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

POST /playlists/{playlistId}/items Insert Playlist Item



Adds a video to the specified playlist.

Parameters

Try it out

Name	Description
playlistId * required string (path)	The unique ID of the playlist.

playlistId

Request body required

application/json ▾

Example Value | Schema

```
{
  "id": "pli_abc123",
  "snippet": {
    "playlistId": "PL1234567890",
    "position": 0,
    "resourceId": {
      "videoId": "dQw4w9WgXcQ"
    }
  }
}
```

Responses

Code	Description	Links
201	Item successfully added to the playlist. Media type application/json Controls Accept header. Example Value Schema <pre>{ "id": "pli_abc123", "snippet": { "playlistId": "PL1234567890", "position": 0, "resourceId": { "videoId": "dQw4w9WgXcQ" } } }</pre>	No links
400	Bad Request — invalid parameters or payload. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	No links
401	Unauthorized — missing or invalid Bearer token. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>	No links

PUT

/playlists/{playlistId}/items/{itemId} Update Playlist Item



Modifies a playlist item, e.g. updating its position.

Parameters

Try it out

Name	Description
playlistId * required string (path)	The unique ID of the playlist.
itemId * required string (path)	The unique ID of the playlist item.

Request body **required**

application/json

Example Value | Schema

```
{
  "id": "pli_abc123",
  "snippet": {
    "playlistId": "PL1234567890",
    "position": 0,
    "resourceId": {
      "videoId": "dQw4w9WgXcQ"
    }
  }
}
```

Responses

Code	Description	Links
200	Playlist item successfully updated.	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "id": "pli_abc123",
  "snippet": {
    "playlistId": "PL1234567890",
    "position": 0,
    "resourceId": {
      "videoId": "dQw4w9WgXcQ"
    }
  }
}
```

401	Unauthorized — missing or invalid Bearer token.	No links
-----	---	----------

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

404	Not Found — resource does not exist.	No links
-----	--------------------------------------	----------

Media type

application/json

Example Value | Schema

```
{
  "error": {
    "code": 400,
    "message": "Invalid Request"
  }
}
```

DELETE /playlists/{playlistId}/items/{itemId} Delete Playlist Item



Removes a video from the playlist.

Parameters

Try it out

Name	Description
playlistId * required string (path)	The unique ID of the playlist.
itemId * required	The unique ID of the playlist item.

Name	Description
string (path)	itemId

Responses

Code	Description	Links
204	Playlist item successfully deleted.	No links
401	Unauthorized — missing or invalid Bearer token. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links
404	Not Found — resource does not exist. Media type application/json Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>	No links

Module 3 ▶ Subscriptions

Module: User Preference & Personalization Control

User channel subscription management — list, subscribe, and unsubscribe. Bell-icon alert status is checked internally before dispatching Type 1 push notifications.

^

GET

/subscriptions

List Subscriptions



Retrieves a list of active channel subscriptions for the authenticated user.

Parameters

No parameters

Try it out

Responses

Code	Description	Links
200	Subscriptions retrieved successfully. Media type application/json Controls Accept header. Example Value Schema <pre>{ "nextPageToken": "CAIQAA",</pre>	No links

Code	Description	Links
401	<pre>"items": [{ "id": "su_1234567890" }]</pre> Unauthorized — missing or invalid Bearer token.	No links
Media type application/json		
Example Value Schema		
<pre>{ "error": { "code": 400, "message": "Invalid Request" } }</pre>		

POST /subscriptions Insert Subscription

Adds a new channel subscription to the user's profile.

Parameters

No parameters

Request body required

application/json

Example Value | Schema

```
{  
  "channelId": "UC_x5XG10V2P6uZZ5FSM9Ttw"  
}
```

Responses

Code	Description	Links
201	Subscribed successfully.	No links

DELETE /subscriptions/{channelId} Delete Subscription

Removes a specific channel from the user's subscriptions.

Parameters

Name	Description
channelId * required string (path)	channelId

Responses

Code	Description	Links
204	Unsubscribed successfully.	No links

Module 3 ▶ Inter-Service: Data Services

Module: User Preference & Personalization Control — INTERNAL

Core database mutation and retrieval APIs used exclusively by internal microservices. All routes prefixed `/internal/`. Blocked from external access at the gateway.

GET

/internal/users/{userId}/context

Get User Context (AI Engine → Data Services)

Called by the AI Engine to fetch aggregated User Profile, Preferences, and Watch History to formulate the Personalization Strategy.

Parameters

Try it out

Name	Description
userId * required string (path)	userId

Responses

Code	Description	Links
200	Complete user context.	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{
  "profile": {
    "privacy": {
      "dataTrackingEnabled": true
    },
    "prefVector": {
      "interests": [
        "music",
        "gaming"
      ]
    }
  },
  "recentHistoryIds": [
    "string"
  ]
}
```

PUT

/internal/users/{userId}/profile

Sync Settings (User Control Center → Data Services)

Cascades user settings updates down to the core database.

Parameters

Try it out

Name	Description
userId * required string (path)	userId

Request body required

application/json ▾

Example Value | Schema

```
{
  "privacy": {
    "dataTrackingEnabled": true
  },
  "prefVector": {
    "interests": [
      "music",
      "gaming"
    ]
  }
}
```

Responses

Code	Description	Links
200	Core profile updated.	No links

GET

/internal/users/{userId}/subscriptions/{channelId}/status Get Subscription Alert Status (Notification → Data Services)



Checks if the user has the bell icon enabled for a specific channel before sending a Type 1 push.

Parameters

Try it out

NameDescription

userId ★ required
string
(path)

userId

channelId ★ required
string
(path)

channelId

Responses

Code	Description	Links
200	Subscription alert status.	No links

Media type

application/json ▾

Controls Accept header.

Example Value | Schema

```
{
  "alertsEnabled": true
}
```

PUT

/internal/videos/{videoId}/trending-score Update Trending Score (Analytics/Feedback → Data Services)



Mutates the video's trending score based on explicit feedback, shares, or completed watch sessions.

Parameters

Try it out

Name	Description
videoid * required	
string (path)	videoid

Request body required

application/json

Example Value | Schema

```
{
  "scoreDelta": 2.5,
  "source": "FeedbackCollector"
}
```

Responses

Code	Description	Links
200	Trending score updated.	No links

Module 4 ▶ Notifications

Module: Notification System

Client-facing notification inbox (list, mark as read) and push preference management (global enable/disable toggle).

^

GET

/notifications

List Notifications (Inbox)

🔒 ^

Retrieves the authenticated user's recent notification inbox.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Array of notification payloads. Media type application/json Controls Accept header. Example Value Schema <pre>[{ "notificationId": "notif_789", "notificationType": "NEW_UPLOAD", "status": "DELIVERED" }]</pre>	No links
401	Unauthorized — missing or invalid Bearer token. Media type application/json	No links

Code	Description	Links
Example Value Schema <pre>{ "error": { "code": 400, "message": "Invalid Request" }}</pre>		

GET

/notifications/preferences

Get Push Preferences

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Returns push preference settings. Media type application/json Controls Accept header. Example Value Schema <pre>{ "pushEnabled": true}</pre>	No links

PUT

/notifications/preferences

Update Push Preferences

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{  "pushEnabled": true}
```

Responses

Code	Description	Links
200	Notification preferences successfully updated.	No links

PUT

/notifications/{notificationId}/read

Mark Notification as Read

Parameters

Try it out

Name	Description	
notificationId ★ required string <i>(path)</i>	notificationId	
Responses		
Code	Description	Links
204	Notification successfully marked as read.	No links

Module: Notification System — INTERNAL

Module 4 ▶ Inter-Service: Event Triggers

Async event webhooks that wake the Notification Manager. All routes prefixed `/internal/`. **Type 1** — new upload trigger (Ingestion → Notification Manager → subscribers). **Type 2** — AI relevance trigger (AI Engine → Notification Manager → target user). Also covers delivery logging to prevent duplicate pushes.

^

POST

/internal/notifications/trigger/new-upload

Trigger New Upload Notification (Ingestion → Notification)

^

Wakes up the Notification Manager to fan out Type 1 push notifications to all bell-enabled subscribers of the channel.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value

Schema

```
{
  "channelId": "c456",
  "videoId": "v123"
}
```

Responses

Code	Description	Links
202	Event accepted for async processing.	No links

POST

/internal/notifications/trigger/relevance-detected

Trigger AI Relevance Push (AI Engine → Notification)

^

Tells the Notification Manager a high-confidence video match was found — initiates a Type 2 recommendation push.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "userId": "u001",
  "videoId": "v123",
  "relevanceScore": 0.98
}
```

Responses

Code	Description	Links
202	Recommendation push queued.	No links

POST

/internal/notifications/log Log Notification Delivery (Notification → Data Services)

^

Records that a notification was dispatched so the user isn't spammed with duplicates.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "notificationId": "notif_789",
  "notificationType": "NEW_UPLOAD",
  "status": "DELIVERED"
}
```

Responses

Code	Description	Links
201	Delivery log saved.	No links

Schemas ^

GoogleAuthRequest {

idToken* > [...]

}

AuthResponse {

userId > [...]

accessToken > [...]

refreshToken > [...]

expiresIn > [...]

}

UserProfile ▾ {
 userId > [...]
 email > [...]
 displayName > [...]
 avatarUrl > [...]
 createdAt > [...]
}

SessionCreateRequest ▾ {
 deviceType > [...]
 appVersion > [...]
}

SessionContext ▾ {
 sessionId > [...]
 geoContext > [...]
}

PageInfo ▾ {
 totalResults > [...]
 resultsPerPage > [...]
}

VideoListResponse ▾ {
 nextPageToken > [...]
 pageInfo **PageInfo** > {...}
 items > [...]
}

Video ▾ {
 id > [...]
 snippet > {...}
 statistics > {...}
}

VideoStats ▾ {
 description: Full engagement statistics for a video. Like/dislike counts are authoritative here; Video.statistics carries only viewCount and trendingScore for feed rendering.

 videoId > [...]
 likeCount > [...]
 dislikeCount > [...]
 commentCount > [...]
 shareCount > [...]
}

VideoInteractionRequest ▾ {
 action* > [...]
 playlistId > [...]
 clipSettings > {...}
}

```
StreamManifestResponse ▾ {  
  hlsManifestUrl      > [...]  
  resolutions         > [...]  
}
```

```
ChapterListResponse ▾ {  
  chapters             > [...]  
}
```

```
Comment ▾ {  
  commentId           > [...]  
  authorDisplayName   > [...]  
  text                > [...]  
  publishedAt         > [...]  
  likeCount           > [...]  
}
```

```
CommentListResponse ▾ {  
  nextPageToken       > [...]  
  items               > [...]  
}
```

```
Playlist ▾ {  
  id                  > [...]  
  snippet             > {...}  
  status              > {...}  
}
```

```
PlaylistListResponse ▾ {  
  nextPageToken       > [...]  
  pageInfo            PageInfo > {...}  
  items               > [...]  
}
```

```
PlaylistItem ▾ {  
  id                  > [...]  
  snippet             > {...}  
}
```

```
PlaylistItemListResponse ▾ {  
  nextPageToken       > [...]  
  pageInfo            PageInfo > {...}  
  items               > [...]  
}
```

```
SubscriptionListResponse ▾ {  
  nextPageToken       > [...]  
  items               > [...]  
}
```

PlaybackEventBatch ▾ {
 videoId > [...]
 sessionId > [...]
 events > [...]
}

WatchSummary ▾ {
 sessionId > [...]
 watchMinutes > [...]
 skipRatio > [...]
}

UserSettings >

NotificationLog ▾ {
 notificationId > [...]
 notificationType > [...]
 status > [...]
}

NotificationPreferences ▾ {
 pushEnabled > [...]
}

ErrorProto ▾ {
 error > {...}
}